

Computability & Logic – Re-Exam August 2022

s-number

Name

You have 3 hours to finish this exam. **Note:** this is an open-book exam.

You may use books, lecture slides and other notes on paper.

The use of electronic devices is *not* permitted (dispensations may apply).

Each of the following 30 questions has 4 possible answers of which exactly one is correct. For each question, you may select one or more answers by checking the appropriate boxes. Your test is graded as follows:

- If you select only the correct answer, you receive 2 points.
- If you select 2 answers, one of which is correct, you receive 1 point.
- If you select 3 answers, one of which is correct, you receive 0.41 points.
- If you select no answer or all answers, you receive 0 points.
- If you select only one answer, and it is wrong, you receive -0.67 points.
- If you select 2 answers, that are both wrong, you receive -1 point.
- If you select 3 answers, that are all wrong, you receive -1.25 points.

Note that perfect answers yield a score of 60 and random guessing may give you a negative score. For the logical basis of this score system, see Frandsen and Schwartzbach, *A singular choice for multiple choice*.

Success!

Regular Languages

Question 1

Consider the regular expression $E := (a^*c + b^*c)^*$. Which one of the following words is *not* a member of the language $\mathcal{L}(E)$?

- a ☐ $abac$
- b ☐ $acbc$
- c ☐ $bcac$
- d ☐ $acac$

Question 2

Let $\mathcal{L}_1$, $\mathcal{L}_2$ and $\mathcal{L}_3$ be languages over the same alphabet Σ .

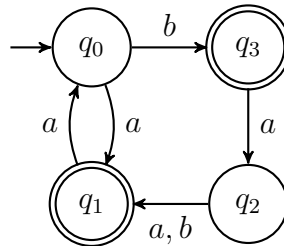
Assume that $\mathcal{L}_1 \subseteq \mathcal{L}_2 \subseteq \mathcal{L}_3$. Assume that $\mathcal{L}_1$ and $\mathcal{L}_3$ are regular.

Which one of the following statements follows from the assumptions above?

- a ☐ If $\mathcal{L}_2$ is regular then $\mathcal{L}_2' \cap \mathcal{L}_3$ is regular.
- b ☐ If $\mathcal{L}_1$ is finite then $\mathcal{L}_2$ is regular.
- c ☐ $\mathcal{L}_2$ is regular.
- d ☐ If $\mathcal{L}_2$ is regular then $\mathcal{L}_3$ is finite.

Question 3

Consider the following *incomplete* DFA M :



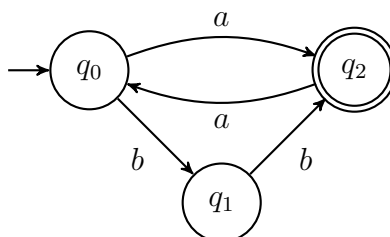
We compute the (unique) minimal *complete* DFA M' , such that $\mathcal{L}(M) = \mathcal{L}(M')$.

What is the number of states of M' ?

- a ☐ 3 states
- b ☐ 2 states
- c ☐ 4 states
- d ☐ 5 states

Question 4

Consider the following incomplete DFA M :

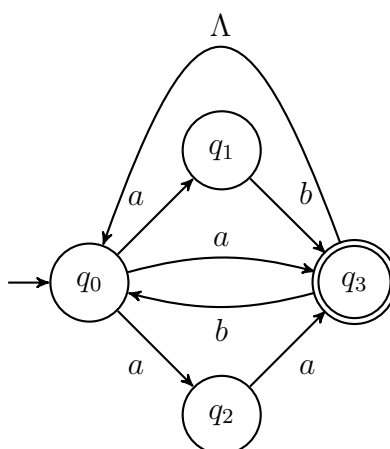


Which of the regular expressions describes the language $\mathcal{L}(M)$?

- a ☐ $(a + bb)(a(a + bb))^*$
- b ☐ $(a + bb)((aa)^*abb)^*$
- c ☐ $(a + bb)((abb)^*aa)^*$
- d ☐ $(a + bb)((aa)^* + (abb)^*)$

Question 5

Consider the following NFA M :



We compute the minimal complete DFA M' , such that $\mathcal{L}(M) = \mathcal{L}(M')$.

What is the number of (accepting) states in M' ?

- a ☐ M' has 4 states, of which 2 states are accepting.
- b ☐ M' has 4 states, of which 1 state is accepting.
- c ☐ M' has 3 states, of which 1 state is accepting.
- d ☐ M' has 3 states, of which 2 states are accepting.

Question 6

We define the following three languages over the alphabet $\Sigma = \{a, b, c\}$. Recall that $\#_a(w)$ is defined as the number of a 's that occur in word w .

- $\mathcal{L}_1 = \{w \in \Sigma^* \mid \#_a(w) \text{ is a multiple of } 5\}$
- $\mathcal{L}_2 = \{w \in \Sigma^* \mid \#_a(w) + \#_b(w) \leq 5\}$
- $\mathcal{L}_3 = \{w \in \Sigma^* \mid \#_a(w) = \#_b(w)\}$

Which of these languages is *regular*?

- ☐ $\mathcal{L}_1$ and $\mathcal{L}_2$ are regular, but $\mathcal{L}_3$ is not regular
 - ☐ None of these languages is regular
 - ☐ Only $\mathcal{L}_1$ is regular; $\mathcal{L}_2$ and $\mathcal{L}_3$ are not regular
 - ☐ $\mathcal{L}_1$ and $\mathcal{L}_3$ are regular, but $\mathcal{L}_2$ is not regular
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Context-Free Languages

Question 7

Consider the following Context-Free Grammar:

$$G = \begin{cases} S & \rightarrow a S a \mid B a \\ B & \rightarrow B B \mid b \end{cases}$$

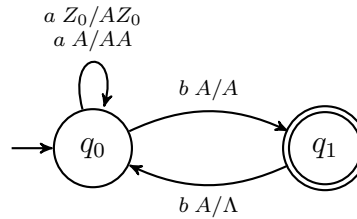
G happens to be an ambiguous grammar.

Which of the following words has more than one derivation tree in G ?

- ☐ The word $abbbaa$
- ☐ The word $aabbbaa$
- ☐ The word $abbaa$
- ☐ The word $aabbaa$

Question 8

Consider the following PDA (Push-Down Automaton).



Which of the following words is accepted by final state in this PDA?

- a ☐ The word *aabbb*
 b ☐ The word *aabb*
 c ☐ The word *aabbbb*
 d ☐ The word *aabbbbb*

Question 9

Consider the following grammar in Chomsky Normal Form.

$$G = \begin{cases} S \rightarrow AB \\ A \rightarrow AX \mid a \\ B \rightarrow BB \mid b \\ X \rightarrow a \end{cases}$$

We apply the CYK-parsing algorithm to the input word *aabb*. The matrix is already partially filled.

1	{A, X}	..	..	{S}
2	—	{A, X}	..	..
3	—	—	{B}	..
4	—	—	—	{B}
	1	2	3	4

How many occurrences of *S* will appear in the whole matrix?

- a ☐ 4 occurrences
 b ☐ 2 occurrences
 c ☐ 1 occurrence
 d ☐ 6 occurrences

Question 10

All of the following languages are Context-Free, except one.

Which language is *not* context-free?

- a ☐ $\{a^m b^n c^p \mid m = n \text{ and } n = p\}$
 b ☐ $\{a^m b^n c^p \mid m = n \text{ or } n = p\}$
 c ☐ $\{a^m b^n c^p \mid m + p = n\}$
 d ☐ $\{a^m b^n c^p \mid m + n = p\}$

Question 11

Which of the following statements is *false*?

- ☐ *a* Every Deterministic Context-Free Language is Regular.
- ☐ *b* Every Regular Language is Deterministic Context-Free.
- ☐ *c* Every Deterministic Context-Free Language is Context-Free.
- ☐ *d* The intersection of a Regular Language and a Context-Free Language is always Context-Free.

Question 12

Assume that L_1 is a Regular Language and L_2 is a Context-Free Language.
(recall that L' denotes the complement of L)

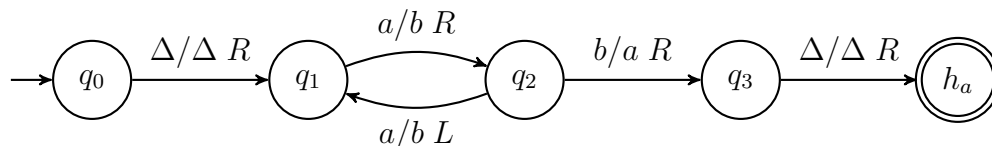
Which of the following statements does *not* follow from these assumptions?

- ☐ *a* L'_2 is Context-Free
- ☐ *b* L'_1 is Context-Free
- ☐ *c* $L_1 \cup L_2$ is Context-Free
- ☐ *d* $L_1 \cap L_2$ Context-Free

Turing Machines and Decidability

Question 13

Consider the following 1-tape Turing Machine on input alphabet $\Sigma = \{a, b\}$:



Which statement on this TM is true?

- ☐ *a* This TM decides the language $\{ab\}$.
- ☐ *b* This TM accepts the language $\{ab\}$ but does not decide it.
- ☐ *c* This TM accepts the language $\{ba\}$ but does not decide it.
- ☐ *d* This TM decides the language $\{ba\}$.

Question 14

Recall that $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ denote the natural, rational, and real numbers, respectively. We order the following sets according to their cardinality. What is the proper order?
(we write $=$ for same cardinality and $<$ for strictly smaller cardinality)

- ☐ $\mathbb{N}^2 = \mathbb{Q}^2 < 2^{\mathbb{N}} = \mathbb{R}$
- ☐ $\mathbb{N}^2 < \mathbb{Q}^2 < 2^{\mathbb{N}} = \mathbb{R}$
- ☐ $\mathbb{N}^2 = 2^{\mathbb{N}} = \mathbb{Q}^2 < \mathbb{R}$
- ☐ $\mathbb{N}^2 < 2^{\mathbb{N}} < \mathbb{Q}^2 = \mathbb{R}$

Question 15

If L is a *recursive* language, then:

- ☐ There exists a deterministic TM that accepts L and terminates on all input.
- ☐ All Turing Machines that accept L terminate on all input.
- ☐ L is generated by a (possibly ambiguous) Context Free Grammar.
- ☐ L' (=complement) cannot be a recursive language.

Question 16

If L is a *recursively enumerable* language, then:

- ☐ There exists a deterministic TM that accepts L and loops on all $w \notin L$.
- ☐ There exists a deterministic TM that accepts L and terminates on all input.
- ☐ L' (=complement) must also be a recursively enumerable language.
- ☐ L' (=complement) cannot be a recursively enumerable language.

Question 17

Which statement is *wrong*?

- ☐ If L_1 and L_2 are recursively enumerable, then so are L'_1 and $L_1 \cap L_2$.
- ☐ If L_1 and L_2 are recursively enumerable, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.
- ☐ If L_1 and L_2 are recursive, then so are L'_1 and $L_1 \cap L_2$.
- ☐ If L_1 and L_2 are recursive, then so are $L_1 \cup L_2$ and $L_1 \cap L_2$.

Question 18

Consider the following decision problems on Turing Machines:

- *Accept- Λ* : Input TM M_1 . Question: $\Lambda \in \mathcal{L}(M_1)$?
- *Equivalent*: Input TM M_2, M_3 . Question: $\mathcal{L}(M_2) = \mathcal{L}(M_3)$?

We may use the following simple Turing Machines as auxiliary components:

- M_Σ : a TM that accepts all words in Σ^* .
- $Write(w)$: a TM that just writes word w on tape and rewinds.
- $Erase$: a TM that erases all input from the tape.

Which of the following problem reductions can be used to show that *Accept- $\Lambda \leq Equivalent$* ? (recall that $\rightarrow$ denotes composition of TMs)

- ☐ a $F(M_1) := (M_\Sigma, Erase \rightarrow M_1)$
- ☐ b $F(M_1) := (Write(\Lambda) \rightarrow M_1, M_\Sigma)$
- ☐ c $F(M_2, M_3) := Write(\Lambda) \rightarrow M_2 \rightarrow M_3$
- ☐ d $F(M_1) := (M_1, Erase \rightarrow M_\Sigma)$

Question 19

Which of the following problems satisfies the two conditions of Rice's theorem?

- ☐ a Given a TM M , is the word Λ accepted by M ?
- ☐ b Given a TM M , does M have an even number of states?
- ☐ c Given a TM M , can M ever write symbol a on tape on input Λ ?
- ☐ d Given a TM M , is the language accepted by M recursively enumerable?

Question 20

Consider the following instance of Post's Correspondence Problem (PCP) with the three pieces $\{(123, 12), (123, 312), (3, 32)\}$. The question is if this is a yes-instance (this instance can be solved).

123	123	3
12	312	32

- ☐ a This instance is not solvable, but other PCP instances can be solved
- ☐ b This instance is solvable, even without reusing any pieces
- ☐ c This instance is solvable, but at least one piece must be reused
- ☐ d This instance is not solvable, and no PCP instances can be solved

Question 21

Which of the following problems on Context-Free Grammars is *undecidable*?

(Σ is the alphabet, and we write L' for the complement of L)

- ☐ a Given input CFG G_1 and G_2 . Question: $\mathcal{L}(G_1)' \cup \mathcal{L}(G_2)' = \Sigma^*$?
 - ☐ b Given input CFG G and word w . Question: $w \in \mathcal{L}(G)$?
 - ☐ c Given input CFG G_1 and G_2 . Question: $\mathcal{L}(G_1) \cup \mathcal{L}(G_2) = \emptyset$?
 - ☐ d Given input CFG G and Regular Expression E . Question: $\mathcal{L}(G) \subseteq \mathcal{L}(E)$?
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Question 22

From the following statements, three are clearly correct, but one is probably wrong (it's status is actually unknown). Which of the following statements is *(probably) not correct*?

- ☐ a If some problem in NP can be solved in polynomial time, then SAT is in P.
 - ☐ b If SAT can be solved in polynomial time, then every NP-complete problem is in P.
 - ☐ c If SAT can be solved in polynomial time, then every problem in NP is in P.
 - ☐ d If some NP-complete problem can be solved in polynomial time, then SAT is in P.
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Logic

Question 23

What is the truth table of the propositional logic formula $(p \rightarrow \neg q) \rightarrow (p \rightarrow q)$?

a ☐	p	q	$(p \rightarrow \neg q) \rightarrow (p \rightarrow q)$
	T	T	T
	T	F	F
	F	T	T
	F	F	T

b ☐	p	q	$(p \rightarrow \neg q) \rightarrow (p \rightarrow q)$
	T	T	F
	T	F	T
	F	T	T
	F	F	T

c ☐	p	q	$(p \rightarrow \neg q) \rightarrow (p \rightarrow q)$
	T	T	F
	T	F	F
	F	T	T
	F	F	T

d ☐	p	q	$(p \rightarrow \neg q) \rightarrow (p \rightarrow q)$
	T	T	F
	T	F	F
	F	T	F
	F	F	F

Question 24

Consider the following formula in propositional logic: $(\neg p \rightarrow p) \rightarrow p$.

This formula is:

- a ☐ A tautology.
 - b ☐ Satisfiable, but not valid.
 - c ☐ Unsatisfiable.
 - d ☐ None of the above.
-

Question 25

Consider the list of proof rules of natural deduction for proposition logic:

$(\text{premise/assumption}), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2, \rightarrow i, \rightarrow e, \neg i, \neg e, \neg\neg e, \perp e$.

The sequent " $(\neg p \rightarrow q) \vdash (\neg q \rightarrow p)$ " can be proved by the following box-style proof, where we replaced the names of the applied rules by question marks:

1.	$\neg p \rightarrow q$	(premise)
2.	$\neg q$	(assumption)
3.	$\neg p$	(assumption)
4.	q	???
5.	$\perp$	???
6.	$\neg\neg p$	???
7.	p	???
8.	$\neg q \rightarrow p$	???

Which rules fit on the five question marks in the proof? (in the correct order)

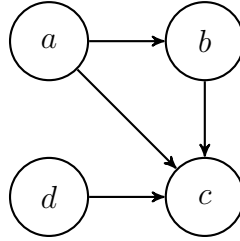
- a ☐ $\rightarrow e, \neg e, \neg i, \neg\neg e, \rightarrow i$
- b ☐ $\rightarrow i, \neg e, \neg i, \neg\neg e, \rightarrow e$
- c ☐ $\rightarrow e, \neg i, \perp e, \neg\neg e, \rightarrow i$
- d ☐ $\neg e, \perp i, \neg i, \neg\neg e, \rightarrow i$

Question 26

Consider the following model of predicate logic, where the vocabulary contains a binary relation R . The model consists of a universe $\{a, b, c, d\}$ and the relation

$$R = \{(a, b), (a, c), (b, c), (d, c)\}$$

The model can be visualized as follows:



Which one of the following formulas in predicate logic holds in this model?

- a ☐ $\forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$
- b ☐ $\forall x \forall y \exists z (R(x, z) \wedge R(y, z))$
- c ☐ $\forall x \forall y \exists z (R(z, x) \wedge R(z, y))$
- d ☐ $\exists z \forall x R(x, z)$

Question 27

Consider the following proof rules for Natural Deduction (without equality and negation):

$$(premise/assumption), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2, \rightarrow i, \rightarrow e, \forall i, \forall e, \exists i, \exists e.$$

Consider the following sequent in Predicate Logic (a and b are constants):

$$R(b) \rightarrow Q(b), \quad \forall x (P(x) \rightarrow (P(a) \wedge R(x))) \quad \vdash \quad P(b) \rightarrow \exists x Q(x)$$

In the statements below, we disregard applications of premise/assumption, and the mentioned rules can be applied any number of times in any order.

Which statement is true?

- a ☐ This sequent can be proved using rules $\rightarrow e, \rightarrow i, \wedge e_2, \forall e, \exists i$ (in any order)
- b ☐ This sequent is not a theorem, so it cannot be proved in natural deduction
- c ☐ This sequent can be proved using rules $\rightarrow e, \rightarrow i, \wedge e_2, \exists e, \forall i$ (in any order)
- d ☐ This sequent can be proved using rules $\rightarrow e, \rightarrow i, \wedge i, \forall e, \exists e$ (in any order)

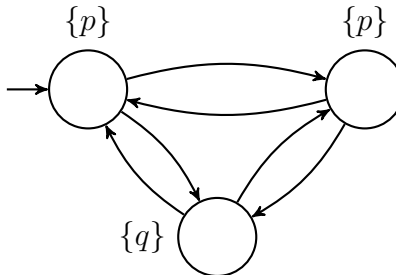
Question 28

Which one of the following statements is *wrong*?

- a ☐ If ϕ is not satisfiable in Arithmetic, then $\neg\phi$ is provable in Peano Arithmetic.
- b ☐ If ϕ is not satisfiable in Predicate Logic, then $\neg\phi$ is provable in Natural Deduction.
- c ☐ If ϕ is provable in Natural Deduction, then $\neg\phi$ is not satisfiable in Predicate Logic.
- d ☐ If ϕ is provable in Peano Arithmetic, then $\neg\phi$ is not satisfiable in Arithmetic.

Question 29

Consider the following Kripke structure:



Which one of the following LTL formulas holds for this Kripke structure?

- a ☐ $p \mathbf{W} q$
- b ☐ $p \mathbf{U} q$
- c ☐ $\mathbf{F} \mathbf{G} q$
- d ☐ $\mathbf{G} (q \rightarrow p)$

Question 30

Consider the following program P , on integer variables x, y, z :

$$\begin{aligned} x &:= y + 1; \\ z &:= x + 1; \\ y &:= z + 1; \end{aligned}$$

Only one of the following Hoare triples about P is *correct*. Which one?

- a ☐ $\{y > 3 \wedge y < 7\} P \{y > 5 \wedge x < 9\}$
- b ☐ $\{y > 5 \wedge y < 9\} P \{y > 5 \wedge x < 9\}$
- c ☐ $\{y > 2 \wedge x < 8\} P \{y > 5 \wedge x < 9\}$
- d ☐ $\{x > 2 \wedge y < 8\} P \{y > 5 \wedge x < 9\}$